Computational biology

Homology and sequence alignment

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Grenoble
Statistiques pour les sciences du Vivant et de l'Homme

September 25, 2026

Today's outline: from gene sequence to protein structure

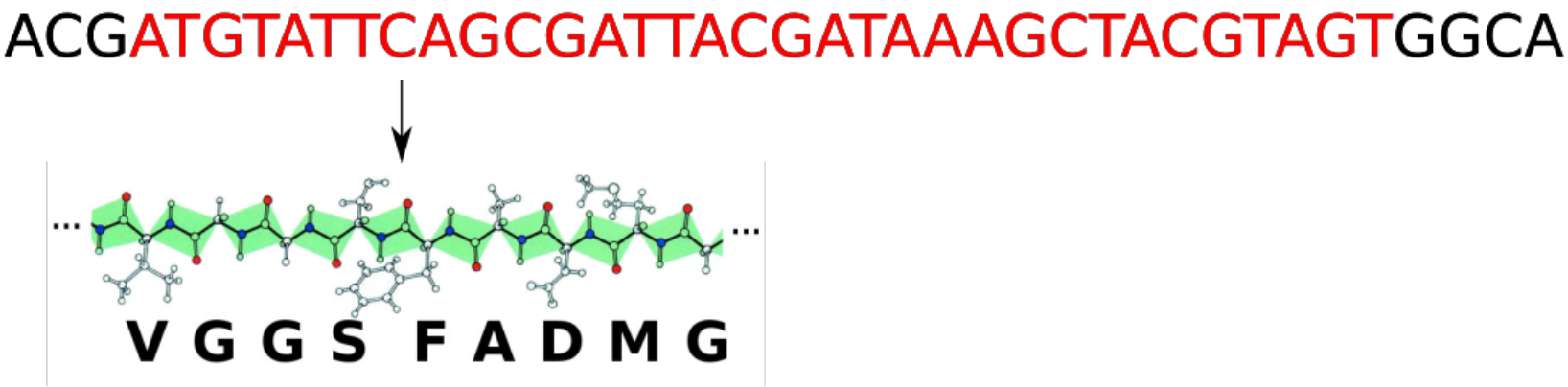
- Sequence-structure-function paradigm
  - Genomes, genes, proteins
  - Databases
- Evolution
  - Selection
  - Sequence homology
  - Multiple sequence alignment

From genome to function, the very big picture

ACGATGTATTCAGCGATTACGATAAAGCTACGTAGTGGCA

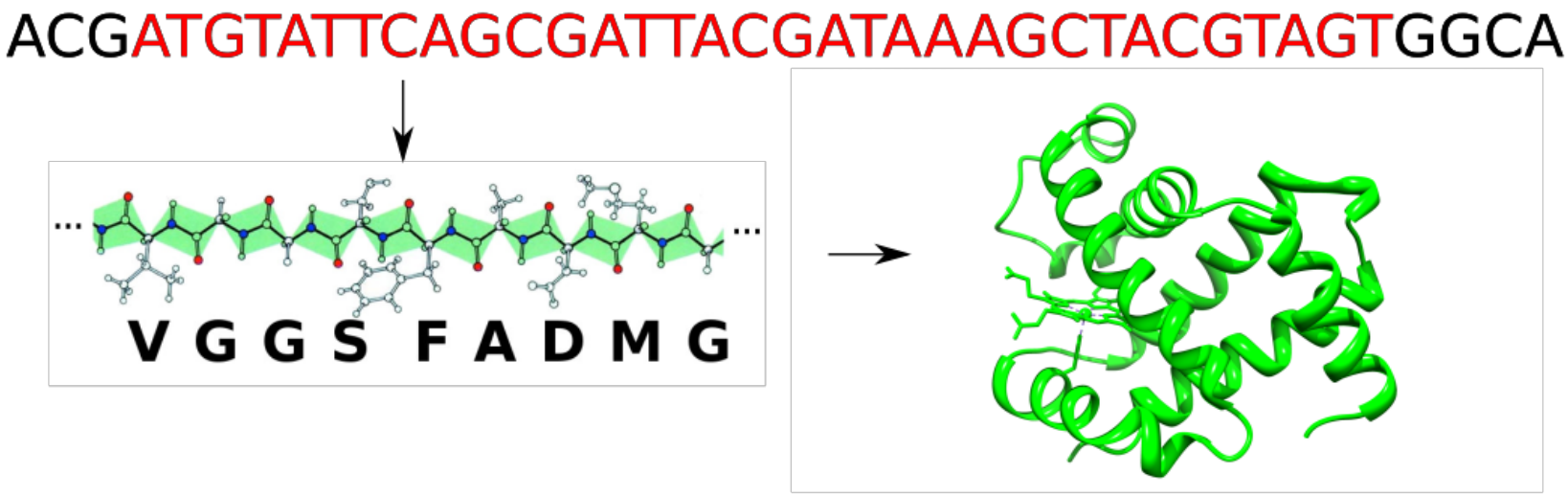
On a genome, specific motifs define begining and end of a gene

From genome to function, the very big picture



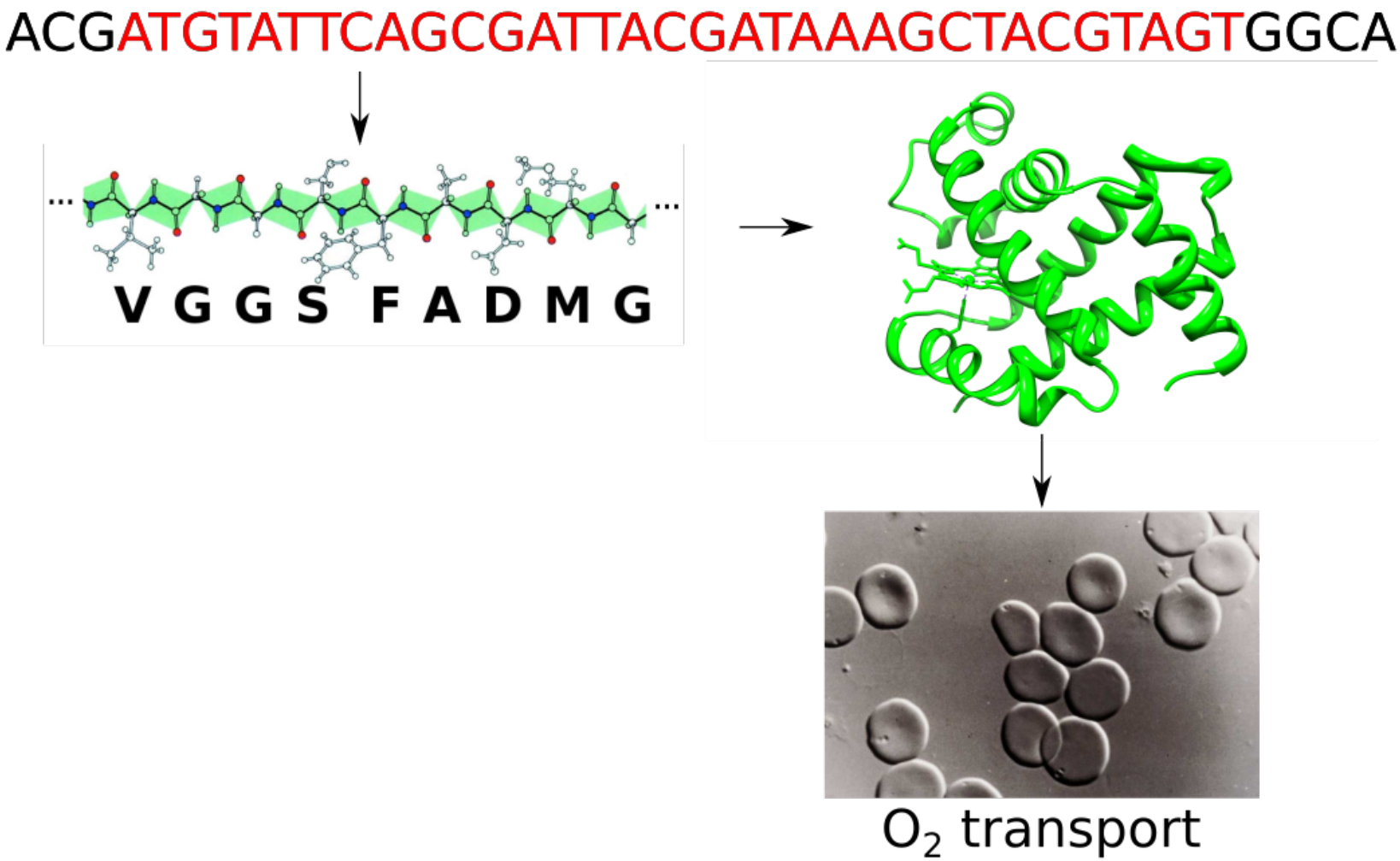
Transcription + *translation*, to form a chain of amino acids (∼300-3000AA)

From genome to function, the very big picture



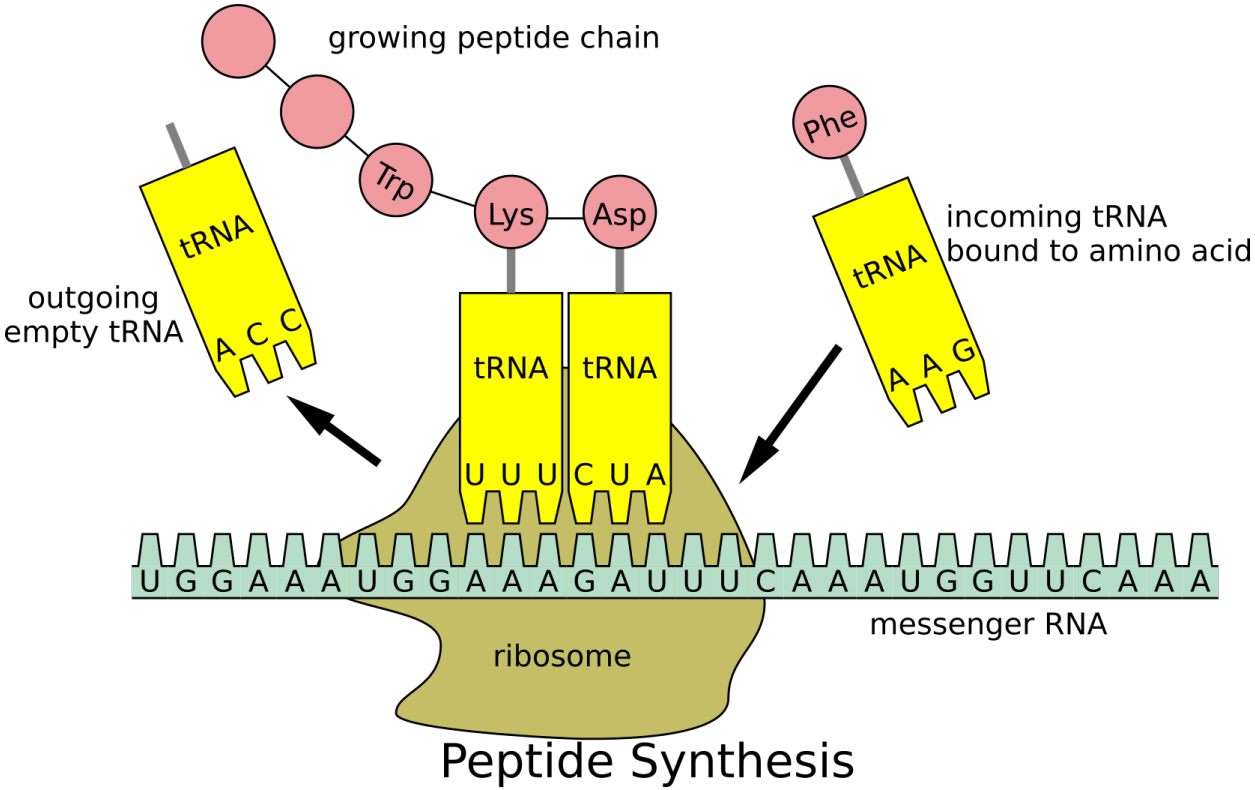
Protein folding under pysico-chemical interactions, diameter $\sim$ few nanometers

From genome to function, the very big picture



Protein endowed with a function (biochemical reactions, transport, etc.)

Zoom: genes to proteins (translation)



RNA codon table

1st position	2nd position U	2nd position C	2nd position A	2nd position G	3rd position
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	stop	stop	A
	Leu	Ser	stop	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

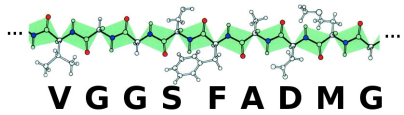
Amino Acids

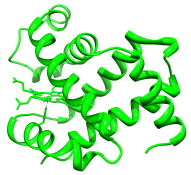
Ala: Alanine
Arg: Arginine
Asn: Asparagine
Asp:Aspartic acid
Cys:Cysteine
Gln: Glutamine
Glu: Glutamic acid
Gly: Glycine
His: Histidine
Ile: Isoleucine
Leu: Leucine
Lys: Lysine
Met: Methionine
Phe: Phenylalanine
Pro: Proline
Ser: Serine
Thr: Threonine
Trp: Tryptophane
Tyr: Tyrosisne
Val: Valine

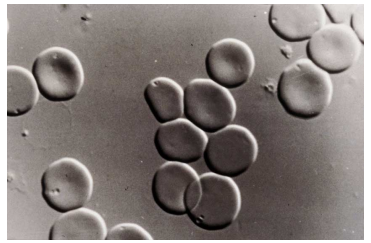
Data at every steps

Nucleic seq. | Amino acid seq. | Protein | Function

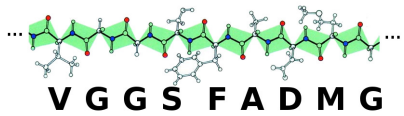
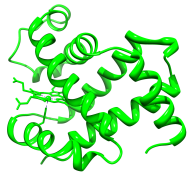
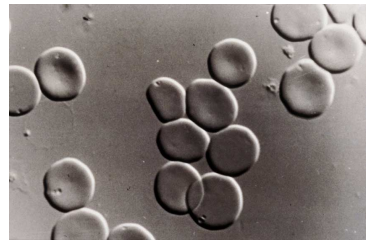
..ATTGTCGATGAC..



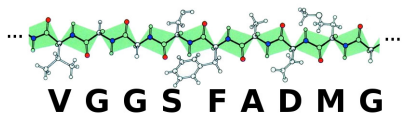
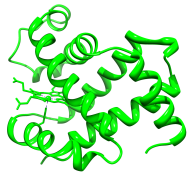
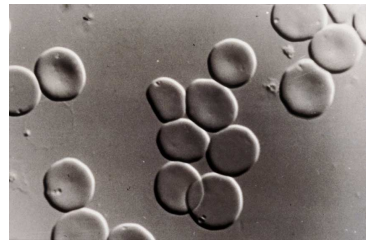




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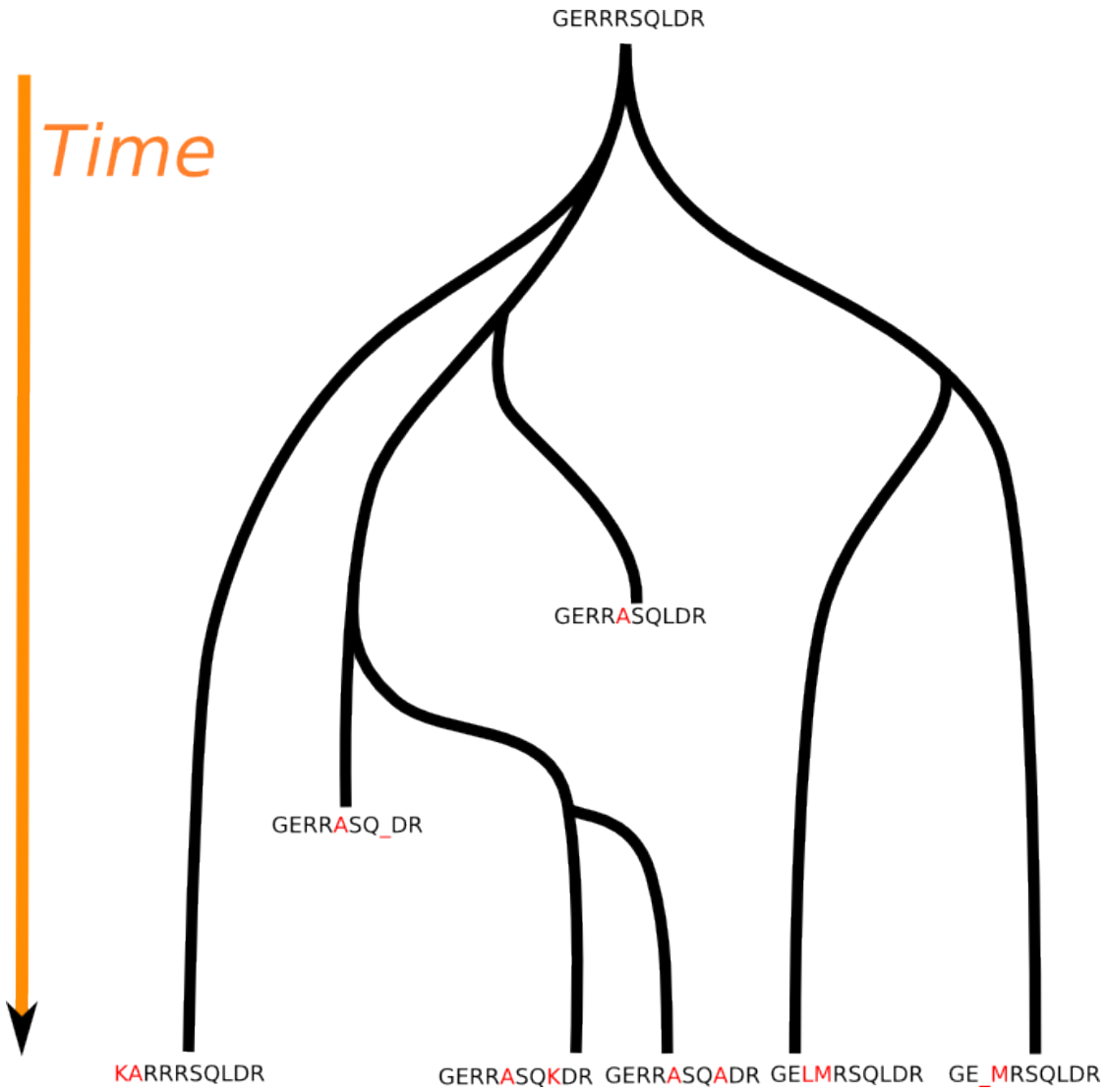
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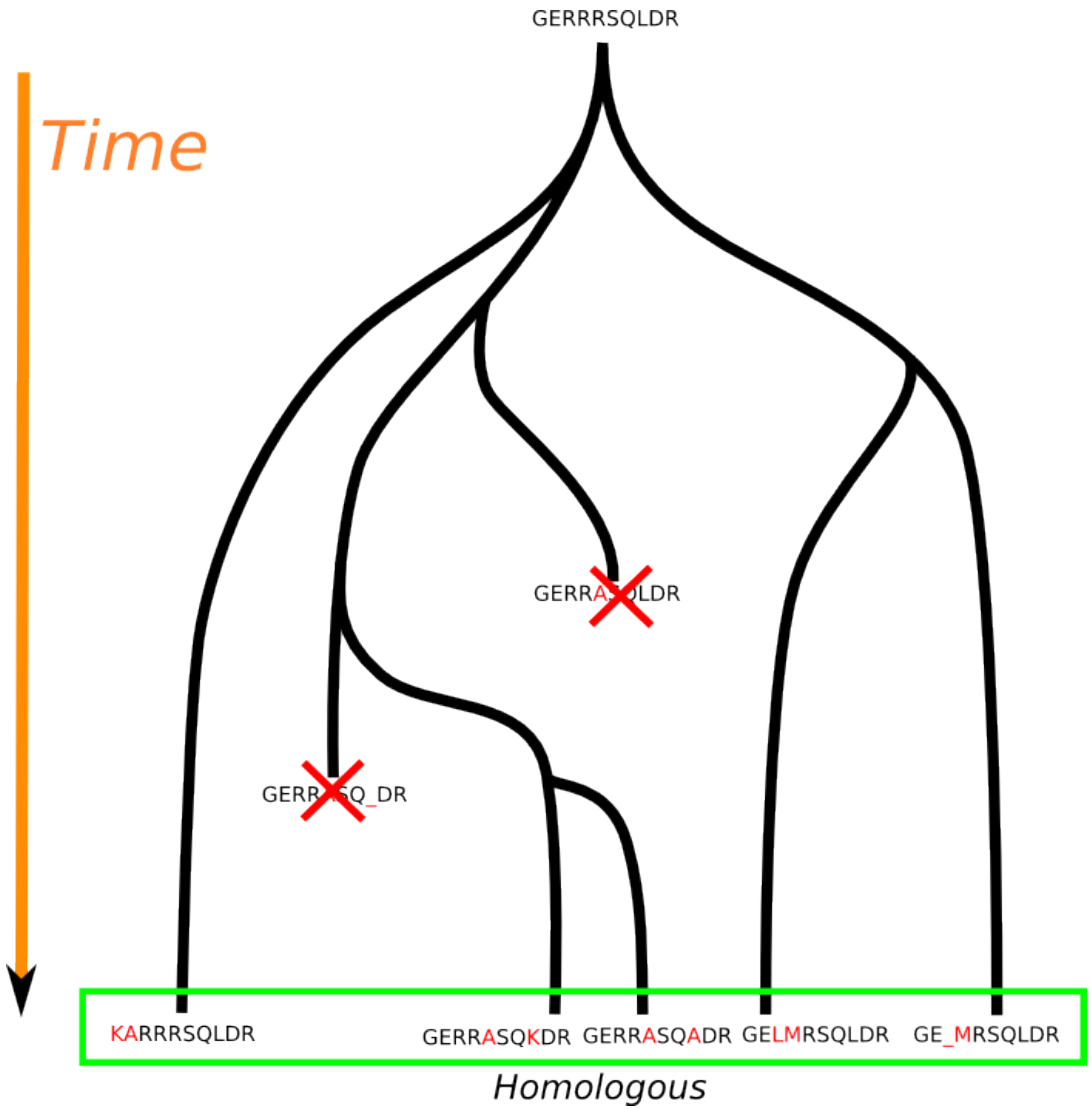
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How do we predict the function from the sequence?

Protein evolution through mutations



Protein evolution through mutations



Smith-Waterman and p-value (blackboard)

Sequence alignement: algorithm and p-value

Find the best alignment between your query sequence S_Q and a reference sequence S_R:

```
MEAIGNA.GSAI
QEAIGNAMGSNI
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Algorithm (sketch):

- given a 20×20 matrix of scores between amino-acids, set gap penalties
- find the alignment maximizing the total score.

Can be solved by **dynamic programming** in $\mathcal{O}(L^2)$ (see *Smith-Waterman algorithm*).

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Under a given p-value threshold we estimate the function to be similar.

Big data: need for heuristic

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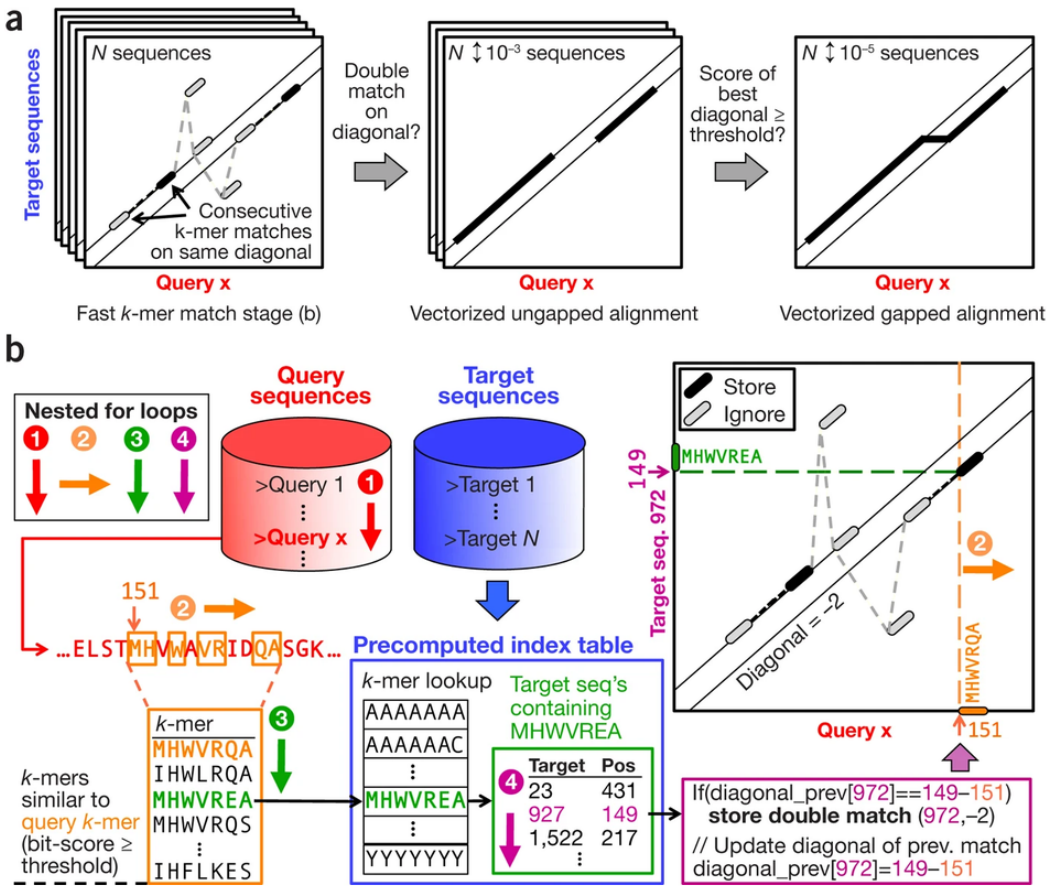
Uniprot (amino acid db) have more than 200M sequences. Even with optimized versions of Smith-Waterman, it is still too heavy to compare sequences to all known sequences.
Tools have developed heuristics to filter down the possible target sequences:

- Blast (the historical tool)
- Diamond
- MMseqs2
- ...

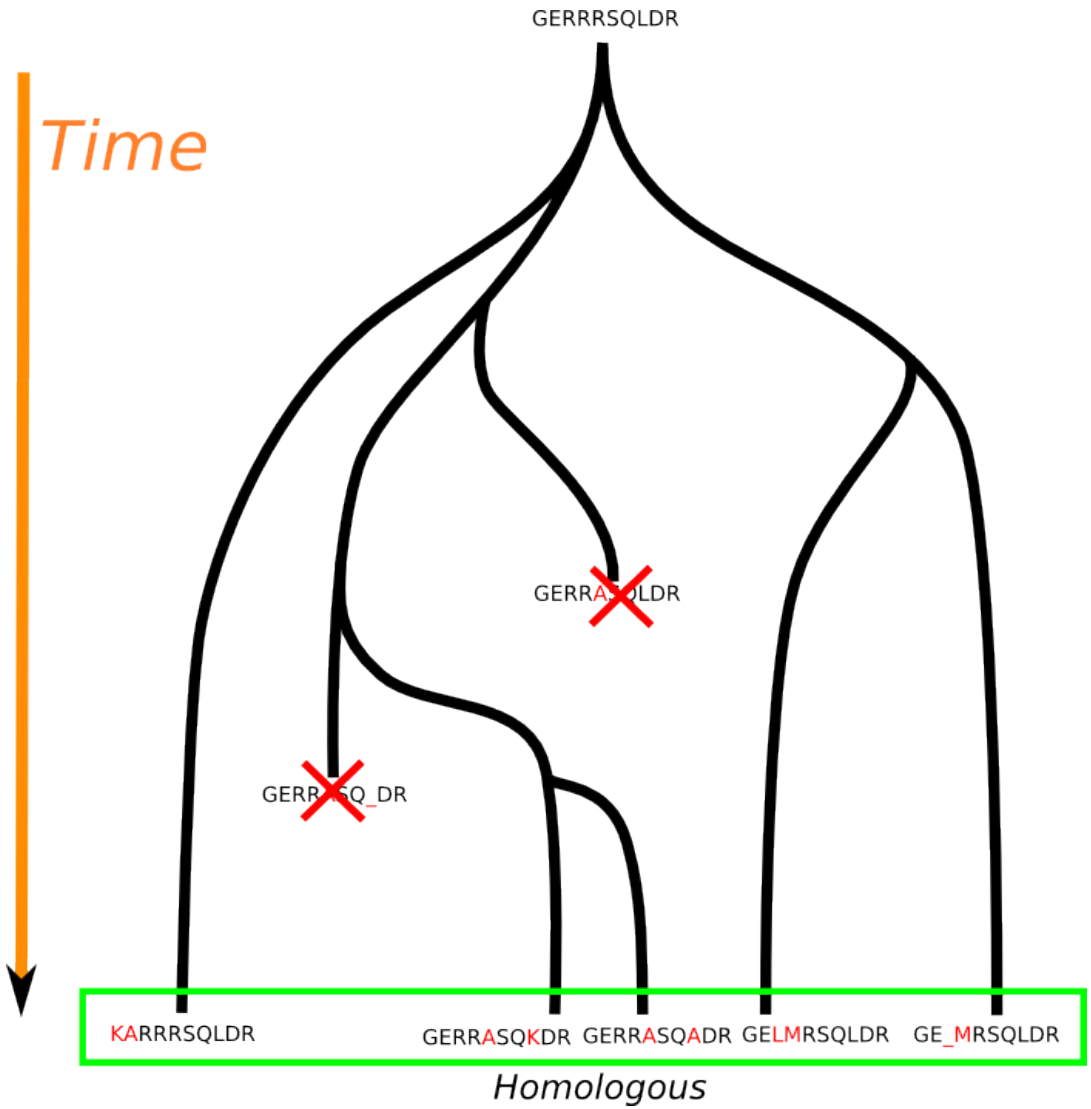
Heuristics are mostly based on efficient pre-filtering (often using similar k-mers, with constant time looks up in hash tables).

MMSeqs2

From: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets



Sequence conservation



Sequence conservation

Aligning the sequences (MSA, multiple sequence alignment):



Tools	Database
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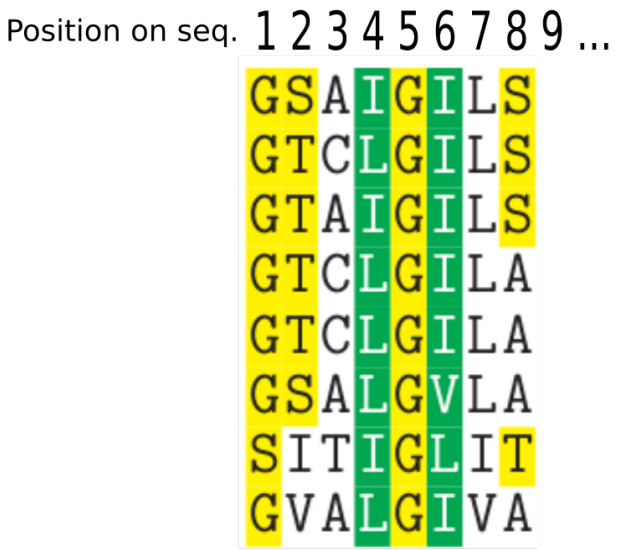
Why some positions are conserved, some other aren't?

Conserved amino acids are essential for the structure/function



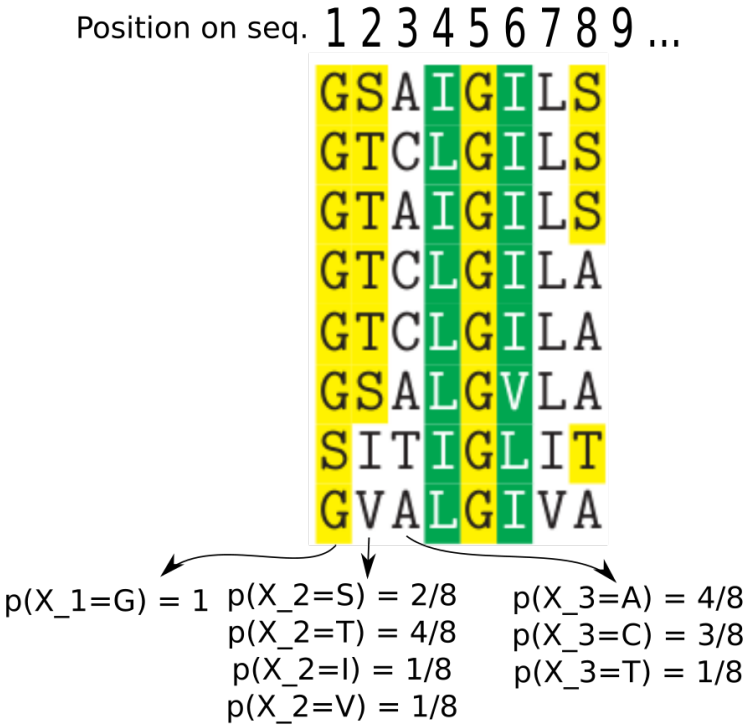
From sequence alignment to profile alignments

From an MSA, we can easily build a **probabilistic model** for modelling the sequence of the protein family:



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By assuming independence of positions, one find the best alignment σ that maximizes the likelihood of a given sequence s=GICLGILA:

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This can be solved again using Smith-Waterman algorithm again.
Matching important (=conserved) positions will play an important role in the likelihood $\rightarrow$ it finds homologs with matching conserved regions.

On-line tools and databases

- Blastn Nucl-Nucl comparison
  `https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastn`
- Blastx Nucl-Prot comparison
  `https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastx`
- Pfam Prot-Prot comparison
  `https://www.ebi.ac.uk/interpro/search/sequence/`
- Protein structure PDB `https://www.rcsb.org/`